

Physical Chemistry: Statistical Thermodynamics

It is not necessary to complete all the exercises, your supervisor will advise you which ones to do; the remainder may be useful for revision. Problems or parts of problems marked with a * are more challenging.

1.

Consider a set of levels with energies 0, 1, 2, 3, 4, 5 and 6 units. Show that for 4 distinguishable particles with a total energy of 6 units the number of microstates, Ω , is 84. If the spacing of the levels is doubled so that their energies are 0, 2, 4 and 6, what is the new value of Ω ?

2.

If an object, with heat capacity C_V , is raised from T_i to T_f at constant volume, the change in entropy, ΔS , is

$$\Delta S = C_V \ln(T_f/T_i).$$

Two identical blocks (heat capacity C_V), one initially at temperature $T + \delta T$ and one at $T - \delta T$, are brought into contact and allowed to come to equilibrium with one another. What will the equilibrium temperature of the two blocks be? Show that the change in entropy of the whole system is given by

$$\Delta S = C_V \ln\left(\frac{T^2}{(T + \delta T)(T - \delta T)}\right)$$

and further show that this expression may be written as

$$\Delta S = -C_V \ln\left(1 - \left(\frac{\delta T}{T}\right)^2\right).$$

Suppose that δT is small; use the approximation that $\ln(1 + x) \approx x$ for $x \ll 1$ to show that ΔS can be approximated by

$$\Delta S = C_V \left(\frac{\delta T}{T}\right)^2.$$

If $T = 300$ K, $\delta T = 0.1$ K and $C_V = 25$ J K⁻¹, what is the probability that, after the two blocks have come to equilibrium, they will subsequently recover their initial temperatures? What value of δT would make this probability 0.001? Comment on your answers.

3.

Consider again the four particles and the set of energy levels described in question 1; let the total energy be 6 units. We will assume that particle 1 is the 'system' and that the remaining three particles form the 'bath'. If the energy of the system, E_m , is 0, show that the number of microstates available to the system plus bath, Ω_m is 28. Repeat the calculation for $E_m = 1, 2 \dots 6$ units.

The probability of finding the system in microstate m , P_m , is given by $\Omega_m/\Omega_{\text{tot}}$, where Ω_{tot} is the total number of microstates of the system plus bath for all energies of the system. A plot of $\ln P_m$ against E_m should be linear (why?). Plot your data in this way (leave off the points for $E_m = 5$ and 6) and comment on the graph.

4.

Consider a system of N non-interacting indistinguishable particles each of which has just two energy levels, one with energy $= 0$ and one with energy $= \Delta$.

- Write down an expression for the partition function, q , of one of these particles. What value does q take in the limits $kT \gg \Delta$ and $kT \ll \Delta$? Make a labelled sketch of how q varies with temperature.
- Use the expression for q from part (a) to find the following expression for the internal energy, U :

$$U = \frac{N\Delta \exp(-\Delta/kT)}{1 + \exp(-\Delta/kT)} = \frac{N\Delta}{\exp(\Delta/kT) + 1}.$$

Show that when $kT \ll \Delta$, U tends to 0, and when $kT \gg \Delta$, U tends to $N\Delta/2$. Comment on these values. Make a *labelled* sketch showing how U varies with temperature.

- Show that the heat capacity of this system is

$$C_V = Nk \left(\frac{\Delta}{kT} \right)^2 \frac{\exp(\Delta/kT)}{[\exp(\Delta/kT) + 1]^2}.$$

By carefully taking the limits show that the heat capacity goes to zero when $kT \gg \Delta$ and when $kT \ll \Delta$. What is the physical interpretation of these results? Make a sketch of how the heat capacity varies with temperature. At roughly what temperature is the heat capacity a maximum?

- *By differentiation, find the temperature at which the heat capacity is a maximum and also find its value at this temperature.

You could make the plots of q , U and C_V using Mathcad or Excel; it is convenient to plot the functions against the dimensionless parameter kT/Δ .

5.

Consider the same system as in question 4, but this time let the lower level have degeneracy g_0 and the upper level have degeneracy g_1 .

- Write down an expression for the partition function, q , of one of these particles. What value does q take in the limit $kT \gg \Delta$ and $kT \ll \Delta$? Make a sketch of how q varies with temperature.
- Use the expression for q from part (a) to derive an expression for U . Find the values of U when $kT \ll \Delta$ and when $kT \gg \Delta$. Comment on the values you obtain. Make a sketch showing how U varies with temperature.
- Without further calculation, make a sketch of how the heat capacity varies with temperature. In what ways do you expect this curve to be different from that in question 4(c)? Give your reasons.

6.

Show that the form of the expression for S given in section 4.3.2 on p. 17 is unaffected by changing the zero of energy i.e. by replacing q by $q' \exp(-\epsilon^0/kT)$.

7.

- (a) Derive the partition function for a particle of mass m confined to a one-dimensional box of length a . Explain carefully any approximations you make and comment on their validity.
- (b) Calculate the thermal wavelength, Λ , for N_2 at 298 K.
- (c) Calculate the molar volume of N_2 gas at 298 K and at a pressure of 1 bar (10^5 N m^{-2}); hence calculate the three-dimensional translational partition function for one mole of N_2 under these conditions. Comment on its value.

8.

- (a) Derive an expression for the rotational partition function of a diatomic molecule, taking the rigid rotor energy levels as a model; you may assume that the energy levels are closely spaced compared to kT . In this context, what is the symmetry number?
- (b) Spectroscopic measurements show that the rotational constant for N_2 is 1.999 cm^{-1} . Find the rotational temperature, θ_{rot} , and hence the value of the rotational partition function at 298 K. Comment on its value in relation to that found in question 7.

9.

- (a) Write down the first few terms of the sum needed to evaluate the vibrational partition function of a diatomic molecule, taking the harmonic oscillator energy levels as a model; you should use a set of energy levels which are measured from the bottom of the potential energy well. Convince yourself that the series you have written down is a geometric progression. Identify the first term and the common ratio, and hence determine the sum to infinity. Write your result in terms of θ_{vib} .
- (b) Spectroscopic measurements show that the vibrational frequency for N_2 is 2360 cm^{-1} . Find the vibrational temperature, θ_{vib} , and hence the value of the vibrational partition function at 298 K: (i) if the energies are measured relative to the bottom of the potential energy well; (ii) if the energies are measured relative to the lowest vibrational level. Comment on these values in relation to one another and to those found in questions 7 & 8.

10.

The energy levels of a Morse oscillator can be written

$$\tilde{E}_v = v\tilde{\omega} - v(v+1)\tilde{\omega}x_e$$

where the energy of the lowest level, $v = 0$, has been taken as zero. Write down the first few terms of the sum that you would need to evaluate in order to find the vibrational partition function.

For N_2 $\tilde{\omega} = 2360 \text{ cm}^{-1}$ and $\tilde{\omega}x_e = 14.5 \text{ cm}^{-1}$; evaluate the above vibrational partition function at 298 K by adding together as many terms as are sufficient to give an answer to three decimal places. Do the same for Br_2 which has $\tilde{\omega} = 323 \text{ cm}^{-1}$ and $\tilde{\omega}x_e = 1.1 \text{ cm}^{-1}$.

Compare these values of the partition functions with those computed assuming the simple harmonic oscillator energy levels (i.e. ignoring the term in x_e). Comment on the results.

11.

- (a) How many normal modes would you expect O_3 to have if its geometry were (i) linear and (ii) bent?
- (b) The three normal modes of H_2O have frequencies 3652, 1595 and 3756 cm^{-1} . Measuring energies from the bottom of the potential energy well, compute the vibrational partition function, at 1000 K, for each mode. Also, compute the total vibrational partition function.

12.

- (a) The following data have been obtained for $\text{Si}(\text{g})$:

level	$^3\text{P}_0$	$^3\text{P}_1$	$^3\text{P}_2$	$^1\text{D}_2$	$^1\text{S}_0$
energy / cm^{-1}	0.0	76	222	6297	15390

Write down the value of the degeneracy, g_j , for each level and hence calculate the value of the electronic partition function of Si at 500 K.

- (b) At room temperature, what are the electronic partition functions of H_2 , O_2 and H_2^+ ?

13.

The electronic ground state of O_2 is $^3\Sigma_g^-$; the first excited state is doubly degenerate and lies 0.97 eV above the ground state. Calculate the electronic partition function at 298 K and at 1000 K (to three decimal places). [Take care with units!]

14.

Given that for N_2 $\tilde{B} = 1.999\text{ cm}^{-1}$ and $\tilde{\omega} = 2360\text{ cm}^{-1}$, what would you expect the molar heat capacity, C_V , to be at 298 K? Explain your reasoning. [Detailed calculations are not necessary]

15.

What would you expect the molar heat capacity, C_V , to be for $\text{H}_2\text{O}(\text{g})$ at 298 K? How would this value be affected if water were, hypothetically, a linear molecule? Explain your reasoning.

16.

In question 4 we considered a system consisting of particles with just two energy levels. In contrast, a system of harmonic oscillators has an infinite number of equally spaced energy levels. Contrast the temperature dependence of q , U and C_V between the two systems, accounting for any differences that you can.

17.

*Consider a system consisting of N particles each of which has three (non-degenerate) energy levels with energies 0, Δ_1 and Δ_2 . Assuming that $\Delta_2 \gg \Delta_1$, make a *labelled* sketch of how you would expect U and C_V to vary with temperature. [Detailed calculations are not required].

18.

- (a) Derive an expression for q_{vib} for a harmonic oscillator, taking the $v = 0$ state to be the energy zero.
- (b) Hence, find expressions for U and C_V .
- (c) Explain what ‘low’ and ‘high’ temperature means in the context of this system and carefully find the values of U and C_V in these two limits.
- (d) Comment on the limiting values you have obtained in (c).

19.

Spectroscopic measurements on $\text{Br}_2(\text{g})$ show that the vibrational frequency is 323 cm^{-1} . Determine θ_{vib} and find the vibrational contribution to the molar internal energy and heat capacity of this molecule at 298 K. Comment on the values you obtain.

20.

The electronic ground state of NO is doubly degenerate; lying at 121.1 cm^{-1} above this is an excited state which is also doubly degenerate. Compute the electronic contribution (per mole) to the internal energy of NO at 1000 K. Express this as a percentage of the total contribution from translation and rotation.

How would you expect the electronic contribution to the heat capacity to vary with temperature?

21.

- (a) The bond length of H^{35}Cl is 127.5 pm. Calculate $\tilde{B}$ (in cm^{-1}) and hence θ_{rot} .
- (b) Compute the rotational partition function at 298 K. What is the rotational contribution to the molar internal energy at this temperature?
- (c) Compute the rotational partition function at 30 K. [You will not be able to use the simple formula for this but will need to compute q_{rot} by adding sufficient terms together]
- (d) Compute dq_{rot}/dT at 30 K. [Again, you will have to do this term by term, see Section 4.6.1 starting on p. 21.
- (e) Hence compute the molar rotational contribution to the internal energy at 30 K.

22.

One simple model for a solid is to assume that the particles execute simple harmonic motion about their (fixed) lattice points; it is supposed that the particles are free to oscillate independently in three-dimensions and that they all have the same frequency, ν_E .

If there are N particles, show that the internal energy due to these vibrations (taking the ground level to have zero energy) is given by

$$U = \frac{3Nk\theta_E}{\exp(\theta_E/T) - 1}$$

where $\theta_E = h\nu_E/k$. Show that the corresponding molar heat capacity is given by

$$C_V = \frac{3N_A k \theta_E^2}{T^2} \frac{\exp(\theta_E/T)}{[\exp(\theta_E/T) - 1]^2}.$$

By carefully taking the limit, determine the value of C_V when $T \gg \theta_E$ [hint $e^x = 1 + x$ for $x \ll 1$]. Compare your answer with Dulong and Petit's Law which states that 'all solids have a molar heat capacity of $3R$ '.

[This is the Einstein model of the heat capacity of a solid; it is broadly in agreement with experiment, but fails to predict the T^3 dependence observed at low temperatures; to obtain this, a more sophisticated model, due to Debye, is needed in which there are a range of frequencies of oscillation rather than a single frequency.]

23.

For O_2 $\theta_{rot} = 2.07$ K and $\theta_{vib} = 2256$ K. Calculate the standard molar entropy of $O_2(g)$ at 298 K. [You will need to consider translational, rotational and electronic contributions; vibrational contributions are negligible, why?]

24.

- At 298 K the standard molar entropy of $He(g)$ is $126.1 \text{ J K}^{-1} \text{ mol}^{-1}$. Given that for noble gases the entropy is purely translational, and that $q_{trans} \propto m^{3/2}$, determine S_m° for $Ar(g)$ at the same temperature. [It is not necessary to do a full calculation].
- Use the same approach to find the translational contribution to the standard molar entropy of $N_2(g)$ at 298 K.
- Given that S_m° for $N_2(g)$ is $191.6 \text{ J K}^{-1} \text{ mol}^{-1}$, obtain an estimate for the bond length of N_2 .

25.

- Suggest why it is that whereas the calculated and calorimetric values for the entropy of CO do not agree, those for OCS do.
- For CH_3D it is found that the calculated entropy exceeds the thermochemical value by $11.5 \text{ J K}^{-1} \text{ mol}^{-1}$. Suggest a reason for this, and explain the value of the residual entropy.

26.

*For a homonuclear diatomic formed from nuclei with spin I we can compute the number of symmetric and anti-symmetric nuclear spin states in the following way. Firstly, we note that there are a total of $(2I + 1)^2$ nuclear spin states. Secondly, we note that there must be $(2I + 1)$ symmetric spin states in which the m_I value is the same for the two nuclei. Thirdly we note that the remaining nuclear spin states must be equally divided between symmetric and anti-symmetric states (why?). Show that this line of argument leads to the result given in the handout.

27.

- Describe carefully the intensity pattern you would expect to see in the rotational Raman spectrum of $^{37}Cl_2(g)$ contrasting it with the spectrum you would expect from $^{37}Cl^{35}Cl$. [The nuclear spins of ^{35}Cl and ^{37}Cl are both $\frac{3}{2}$ and are fermions; the term symbol of the electronic ground state is $^1\Sigma_g^+$.]
- How would the rotational Raman spectrum of the species H_2^- compare with that of H_2 ?

28.

In the rotational Raman spectrum of $^{16}\text{O}_2$ lines are observed at the following displacements (in cm^{-1}) from the laser line: 14.450, 26.010, 37.570

In the rotational Raman spectrum of $^{16}\text{O}^{18}\text{O}$ lines are observed at the following displacements (in cm^{-1}) from the laser line: 8.190, 13.650, 19.110, 24.570

Explain, in as much detail as you can, the appearance of these spectra, determine the rotational constant in each case and show that they are consistent. Find the bond length of O_2 .

29.

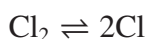
*Show that, in the limit $kT \gg B$,

$$\sum_{\text{even } J} (2J + 1) \exp\left(\frac{-BJ(J + 1)}{kT}\right) = \frac{kT}{2B}$$

[Hint: Let $J = 2K$ and then the required sum runs over $K = 0, 1, \dots$].

30.

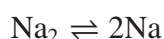
Calculate K_c at 1200 K for the gaseous equilibrium



given the following data: $\theta_{\text{vib}}(\text{Cl}_2) = 808 \text{ K}$, $\theta_{\text{rot}}(\text{Cl}_2) = 0.351 \text{ K}$, $D_e(\text{Cl}_2) = 20,280 \text{ cm}^{-1}$, the electronic ground states of the Cl atoms is $^2\text{P}_{3/2}$ and Cl_2 has a closed shell electronic configuration; take the mass of Cl to be 35 amu. [D_e is the dissociation energy measured from the energy of the atoms to the bottom of the potential energy well; use great care with units.]

31.

- (a) The bond length of Na_2 is 307.8 pm; calculate the moment of inertia, the rotational constant (in cm^{-1}) and the rotational temperature.
- (b) Given that the vibrational frequency of Na_2 is 159.2 cm^{-1} , and that its dissociation energy, D_e , is 0.74 eV, calculate K_p at 1000 K for the equilibrium



32.

- (a) Write down the term symbols for the ground states of Li and Li^+ ; do the same for Cs and Cs^+
- (b) The first ionization energy of Cs(g) is 3.893 eV. Compute the equilibrium constant, K_c , for the ionization of Cs vapour at 2000 K.
- (c) Assume that an initial concentration, c_0 , of Cs atoms is introduced into a vessel and allowed to come to equilibrium according to

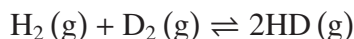


If the degree of ionization is α , show that, provided $\alpha \ll 1$, its value is given by $\alpha = \sqrt{K_c/c_0}$.

Hence determine α for Cs gas at an initial pressure of 0.1 bar and a temperature of 2000 K. Comment on your result.

33.

In the presence of a suitable catalyst, H_2 , D_2 and HD will come to equilibrium according to



The equilibrium constant, K_c , for this reaction can be computed from

$$K_c = \frac{f_{\text{HD}}^2}{f_{\text{H}_2} f_{\text{D}_2}} \exp\left(\frac{-\Delta\epsilon^0}{kT}\right).$$

All of the species H_2 , D_2 and HD have the same potential energy curve (why?) and so $\Delta\epsilon^0 = 0$. For H_2 $\theta_{\text{rot}} = 85.3$ K and $\theta_{\text{vib}} = 6215$ K; the electronic ground state is a singlet.

- (a) Compute the reduced masses (in amu) of H_2 , D_2 and HD .
- (b) Using appropriate ratios of these reduced masses, compute both θ_{rot} and θ_{vib} for D_2 and for HD .
- (c) Compute

$$\frac{f_{\text{trans,HD}}^2}{f_{\text{trans,H}_2} f_{\text{trans,D}_2}}$$

[Note that in this ratio, all the terms apart from the masses cancel]

- (d) Using the values of θ_{rot} from part (b), compute (at 800 K)

$$\frac{q_{\text{rot,HD}}^2}{q_{\text{rot,H}_2} q_{\text{rot,D}_2}}.$$

- (e) Using the values of θ_{vib} from part (b), compute q_{vib} for each of the species and hence find (at 800 K)

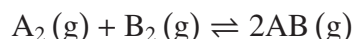
$$\frac{q_{\text{vib,HD}}^2}{q_{\text{vib,H}_2} q_{\text{vib,D}_2}}$$

[You must use the form of q_{vib} in which the energy levels are measured relative to the bottom of the potential energy well]

- (f) Using your results from (c), (d) and (e) compute K_c for the isotopic exchange equilibrium at 800 K. Comment on your result.

34.

Consider the isotopic exchange equilibrium in which A and B are isotopes of the same element:



Assume that the masses of A and B are large enough that the two isotopes can be assumed to have the same mass; this also implies that θ_{rot} and θ_{vib} will be the same for all the diatomics. Using these assumptions compute K_c for this equilibrium. Comment on the result.

35.

In this exercise you will derive the Langmuir isotherm using a simple kinetic argument.

- (a) Assume that there are M surface sites, of which N are occupied by molecules. Explain why the rate of adsorption of molecules onto the surface is given by

$$\text{rate of adsorption} = k_a \times (M - N) \times p,$$

where k_a is a rate constant and p is the pressure of gas. Similarly explain why the rate of desorption from the surface can be written as

$$\text{rate of desorption} = k_d \times N,$$

where k_d is another rate constant.

At equilibrium these two rates are equal. Use this condition to show that

$$\frac{k_a}{k_d} = \frac{\theta}{1 - \theta} \times \frac{1}{p},$$

where θ is the fractional coverage of the surface. Rearrange this to give the following expression for θ :

$$\theta = \frac{K_{\text{ads}} p}{1 + K_{\text{ads}} p},$$

hence determine an expression for K_{ads} in terms of the rate constants. Comment on your answer.

- (b) Assuming that you have measurements of θ as a function of p , determine a straight line plot which can be used to test the Langmuir isotherm and hence determine a value for K_{ads}
- (c) Assuming that k_a and k_d can be written in the Arrhenius form, $A \exp(-E_a/RT)$, determine an expression for K_{ads} in terms of the energy difference between the free and adsorbed molecules, and the pre-exponential factors. Explain why you would only expect there to be significant absorption of the molecules if this process is accompanied by a significant decrease in energy.

36.

- (a) Explain why the number of ways of adsorbing N indistinguishable molecules onto M distinct sites is

$$\frac{M!}{N!(M-N)!}.$$

- (b) Given that

$$Q_N = q^N \times \frac{M!}{N!(M-N)!},$$

show that it follows that

$$A = kT [-N \ln q - M \ln M + N \ln N + (M - N) \ln(M - N)].$$

Hence derive the expression for $\mu(\text{ads})$ given in Eqn. (10.14).

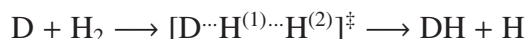
- (c) By equating the chemical potentials of the gaseous and adsorbed gas molecules, derive the expression for θ given in Eqn. (10.15).
- (d) (Harder) Assume that the adsorbed molecule is in fact a closed-shell atom. Explain why it is then reasonable to assume that $q = 1$. Replace μ° in Eqn. (10.15) with the appropriate expression in terms of partition functions (do not forget to take into account the energy difference between the ground electronic states of the free and adsorbed atom). Hence explain why K_{ads} will only be significant if the energy of the adsorbed atom is significantly lower than that of the free atom.

37.

By analogy with the approach adopted in section 11.2 (starting on p. 66) derive an expression for the third order rate constant, k_{3rd} , for the elementary reaction $A + B + C \rightarrow \text{products}$, in which A, B and C come together to form the transition state. Define your terms carefully and show that your final expression has the correct units.

38.

Assuming that the reaction below has a linear transition state, use the data given to compute the rate constant at 1000 K; state the units of your result. [It is advisable to break down the calculation into small parts, such as the separate contributions to the partition functions; be very careful with units].



Reactants: H_2 bond length 0.741 Å, vibrational frequency 4401 cm^{-1} .

Transition state: bond lengths $D-H^{(1)}$ 0.929 Å, $H^{(1)}-H^{(2)}$ 0.929 Å; vibrational frequencies 1708 cm^{-1} (symmetric stretch) and 861 cm^{-1} (bend, doubly degenerate). $\Delta\epsilon_0^\ddagger = 40.3 \text{ kJ mol}^{-1}$.

39.

Use transition state theory to derive an expression for the second order rate constant for a reaction involving two atoms, A and B. Let the masses of the atoms be m_A and m_B , the bond length in the transition state be R , and simply leave the energy separation between the reactants and transition state as $\Delta\epsilon_0^\ddagger$.

Compare your expression with that for the rate constant derived using simple collision theory (see the KCR section 3.2). How can the two expressions be reconciled?

40.

Using the same approach as in Example 16, estimate the steric factor for a reaction between A and BC proceeding via a *non-linear* transition state. Compare your answer with that found for the linear case and comment on the difference.

41.

Using the same approach as in Example 17, estimate the size of the KIE obtained by substituting ^{12}C by ^{13}C . Comment on your answer.

42.

(a) Starting from the Eyring equation

$$k_{2nd} = \left(\frac{1}{c^\circ} \right) \frac{kT}{h} K^\ddagger$$

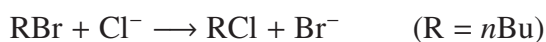
explain how the rate constant can be expressed in terms of an enthalpy and entropy of activation. How are the activation energy, E_a , and the pre-exponential factor, A , in the Arrhenius equation

$$k_{2nd} = A \exp(-E_a/RT)$$

related to the enthalpy and entropy of activation for a reaction in solution?

(b) Describe a graphical method for determining the enthalpy and entropy of activation from the measured temperature dependence rate constants.

(c) Rate constants for the S_N2 reaction (in solution)



as a function of temperature are given in the following table. Use a graphical method to determine the enthalpy and entropy of activation, and the activation energy.

$T/^{\circ}\text{C}$	25.0	34.6	44.5	55.2	64.8
$10^5 \times k_{2\text{nd}} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$	6.45	16.4	41.0	106	215

43.

- (a) Starting from the Eyring equation

$$k_{2\text{nd}} = \left(\frac{1}{c^{\circ}} \right) \frac{kT}{h} K^{\ddagger}$$

explain how measurements of the pressure dependence of the rate constant can be related to the volume of activation of a reaction. Describe a graphical method for determining this volume of activation. Why is the concept of a volume of activation only useful for reactions in solution?

- (b) Rate constant data as a function of pressure for the reactions (in aqueous solution at 298 K)

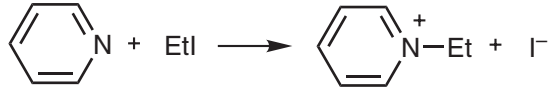


are given below in the table. Use a graphical method to determine the volume of activation in each case.

$p / \text{atmos.}$	10	100	500	1000
$k(p)/k(p = 1)$ for [R1]	1.004	1.041	1.215	1.490
$k(p)/k(p = 1)$ for [R2]	0.9965	0.966	0.839	0.706

44.

Interpret the following data (all reactions are in solution)

reaction	$\Delta_r S^{\circ, \ddagger} / \text{J K}^{-1} \text{ mol}^{-1}$	$\Delta_r V^{\circ, \ddagger} / \text{cm}^3 \text{ mol}^{-1}$
$\text{Cr}(\text{NH}_3)_5(\text{HCONH}_2)^{3+} + \text{H}_2\text{O} \longrightarrow \text{Cr}(\text{NH}_3)_5(\text{H}_2\text{O})^{3+} + \text{HCONH}_2$	-12	-4.8
$\text{Co}(\text{NH}_3)_5(\text{HCONH}_2)^{3+} + \text{H}_2\text{O} \longrightarrow \text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})^{3+} + \text{HCONH}_2$	12	1.1
$\text{Co}(\text{en})_2(\text{OH})(\text{Cl})^+ + \text{H}_2\text{O} \longrightarrow \text{Co}(\text{en})_2(\text{OH})(\text{H}_2\text{O})^{2+} + \text{Cl}^-$	20	
$\text{Pt}(\text{PEt}_3)_2(\text{Cl})(\text{Me})^+ + \text{Br}^- \longrightarrow \text{Pt}(\text{PEt}_3)_2(\text{Br})(\text{Me})^+ + \text{Cl}^-$	-67	
	24	3.2
	-146	-16.8

'en' is the bidentate ligand ethylenediamine $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$; each such molecule provides two N atoms which coordinate to the metal.

45.

- (a) Starting from the bridge relationship, $A = -kT \ln Q_N$, show that, for a system of N non-interacting indistinguishable particles the Helmholtz energy, A , is given by

$$A = -NkT \ln q + kT (N \ln N - N).$$

Explain carefully what q is and how it is defined.

- (b) Given that

$$\left(\frac{\partial A}{\partial V} \right)_T = -p$$

deduce an expression for the pressure of a monotonic gas (i.e. one whose energy levels are just those due to translation). Comment on your result.

- (c) Extend the calculation in (b) to a gas composed of diatomic molecules. Does this affect the pressure?

46.

Calculate the populations at 500 K of the Si atomic levels given in question 12.

47.

- (a) Derive an expression, in terms of B , for the J value of the most populated rotational level at a given temperature.
- (b) For HCl ($\tilde{B} = 10.59 \text{ cm}^{-1}$) it is found in a particular sample that the most populated level is $J = 3$. What is the temperature?
- (c) By how much would the temperature have to rise in order to make $J = 4$ the most populated level?

48.

Consider a sample of $\text{N}_2(\text{g})$ at 298 K and atmospheric pressure held in a container of volume 1 cm^3 . How many translational levels are there with energies on the range 0 to kT ? How does this number compare with the number of particles in the sample? Why is it important that the number of accessible energy levels greatly exceeds the number of particles?

49.

By analogy with the method used for a particle in the three-dimensional box, show that the number of levels with energies in the range 0 to ε for a particle in a two-dimensional box (area A) is given by

$$W_{2D}(\varepsilon) = \frac{2\pi mA\varepsilon}{h^2}.$$

Hence find $D(\varepsilon)$, the density of states.

Molecular energy levels and thermodynamics – summary

Thermodynamics

$$\text{First Law} \quad dU = \delta q + \delta w$$

$$\text{Definition of Helmholtz energy} \quad A = U - TS$$

$$\text{Corresponding Master equation} \quad dA = -p dV - S dT$$

$$\text{hence} \quad \left(\frac{\partial A}{\partial T} \right)_V = -S$$

Definition of partition functions

$$\text{Canonical partition function} \quad Q_N = \sum_{\text{microstates}, m} \exp(-E_m/kT)$$

$$\text{Molecular partition function} \quad q = \sum_{\text{states}, i} \exp(-\varepsilon_i/kT) = \sum_{\text{levels}, j} g_j \exp(-\varepsilon_j/kT)$$

$$\text{separation} \quad q = q_{\text{trans}} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}}$$

$$\text{Indistinguishable particles} \quad Q_N = \frac{q^N}{N!} \quad N! \text{ goes with } q_{\text{trans}}$$

Relation of thermodynamic parameters to partition functions

$$\text{Bridge relationship} \quad A = -kT \ln Q_N$$

$$\text{via } (\partial A / \partial T)_V = -S \quad S = k \ln Q_N + kT \left(\frac{\partial \ln Q_N}{\partial T} \right)_V$$

$$\text{via } A = U - TS \quad U = kT^2 \left(\frac{\partial \ln Q_N}{\partial T} \right)_V$$

In terms of the molecular partition function

$$\text{for all modes} \quad U = NkT^2 \left(\frac{\partial \ln q}{\partial T} \right)_V$$

$$S_{\text{trans}} = Nk \left(\ln q_{\text{trans}} - \ln N + 1 + T \left(\frac{\partial \ln q_{\text{trans}}}{\partial T} \right)_V \right) = Nk (\ln q_{\text{trans}} - \ln N + 1) + \frac{U_{\text{trans}}}{T}$$

$$\text{for rotation, vibration etc.} \quad S_{\text{rot}} = Nk \left(\ln q_{\text{rot}} + T \left(\frac{\partial \ln q_{\text{rot}}}{\partial T} \right)_V \right) = Nk \ln q_{\text{rot}} + \frac{U_{\text{rot}}}{T}$$

Equipartition principle

If the spacing of the energy levels is $\ll kT$, each ‘degree of freedom’ contributes $\frac{1}{2}NkT$ to the internal energy. In molar units, the contribution is $\frac{1}{2}RT$.

Partition functions

$$\text{Translation} \quad E_n = \frac{n^2 h^2}{8ma^2} \quad n = 1, 2, \dots \quad q_{\text{trans,3D}} = \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} V$$

$$\text{Rotation} \quad E_J = BJ(J+1) \quad J = 0, 1, \dots$$

$$B = \frac{\hbar^2}{2I} \quad \tilde{B} = \frac{h}{8\pi^2 c I} \quad \text{diatomic } I = \mu R^2 \quad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$\text{if } kT \gg B \quad q_{\text{rot}} = \frac{kT}{\sigma B} = \frac{T}{\sigma \theta_{\text{rot}}} \quad \theta_{\text{rot}} = \frac{B}{k}$$

$\sigma = 1$ for heteronuclear and $= 2$ for homonuclear diatomics.

Vibration – measuring from the bottom of the PE well

$$E_v = (v + \frac{1}{2})h\nu_0 \quad v = 0, 1, \dots \quad q_{\text{vib}} = \frac{\exp(-h\nu_0/2kT)}{1 - \exp(-h\nu_0/kT)} = \frac{\exp(-\theta_{\text{vib}}/2T)}{1 - \exp(-\theta_{\text{vib}}/T)} \quad \theta_{\text{vib}} = \frac{h\nu_0}{k}$$

Vibration – measuring from the ground state

$$q'_{\text{vib}} = \frac{1}{1 - \exp(-h\nu_0/kT)} = \frac{1}{1 - \exp(-\theta_{\text{vib}}/T)}$$

Nuclear spin statistics

On swapping identical nuclei: Bosons $\Psi \rightarrow +\Psi$; Fermions $\Psi \rightarrow -\Psi$

Symmetry operations to swap nuclei in a homonuclear diatomic: C_2 i_{elec} σ_{elec} p_{nuc}

Number of symmetric nuclear spin states: $(I+1)(2I+1)$,

Number of anti-symmetric nuclear spin states: $I(2I+1)$.

Nuclei with $I = 0, 1, \dots$ are Bosons; those with $I = \frac{1}{2}, \frac{3}{2}, \dots$ are Fermions.

Equilibrium constants

$$\mu_i = -kT \ln \frac{q_i}{N_i} + \varepsilon^{0,i}$$

For the general reaction $\nu_A A + \nu_B B \rightleftharpoons \nu_M M + \nu_N N$

$$K_c = \frac{f_M^{\nu_M} f_N^{\nu_N}}{f_A^{\nu_A} f_B^{\nu_B}} \left(\frac{1}{c^\circ} \right)^{\Delta \nu} \exp(-\Delta \varepsilon_0 / kT)$$

$$\Delta \varepsilon_0 = \nu_M \varepsilon^{0,M} + \nu_N \varepsilon^{0,N} - \nu_A \varepsilon^{0,A} - \nu_B \varepsilon^{0,B} \quad \Delta \nu = \nu_M + \nu_N - \nu_A - \nu_B$$

where f is the volume independent partition function.

$$K_p = \frac{(q_M^\circ)^{\nu_M} (q_N^\circ)^{\nu_N}}{(q_A^\circ)^{\nu_A} (q_B^\circ)^{\nu_B}} (N_A)^{-\Delta \nu} \exp\left(-\frac{\Delta \varepsilon_0}{kT}\right)$$

where q° is the partition function in the standard state (i.e. the volume is the standard molar volume)

Rate constants

For A + BC:

$$k_{\text{2nd}} = \frac{kT}{h} \frac{f'_{\text{TS}}}{f_{\text{A}}f_{\text{BC}}} \exp(-\Delta\varepsilon_0^\ddagger/kT)$$

Thermodynamic formulation (solution) $k_{\text{2nd}} = \frac{1}{[\text{ }]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_{\text{r}}S^{\circ,\ddagger}}{R}\right) \exp\left(\frac{-\Delta_{\text{r}}H^{\circ,\ddagger}}{RT}\right)$

Boltzmann distribution

$$n_i = \frac{N}{q} \exp\left(\frac{-\varepsilon_i}{kT}\right) \quad n_j = \frac{N}{q} g_j \exp\left(\frac{-\varepsilon_j}{kT}\right)$$